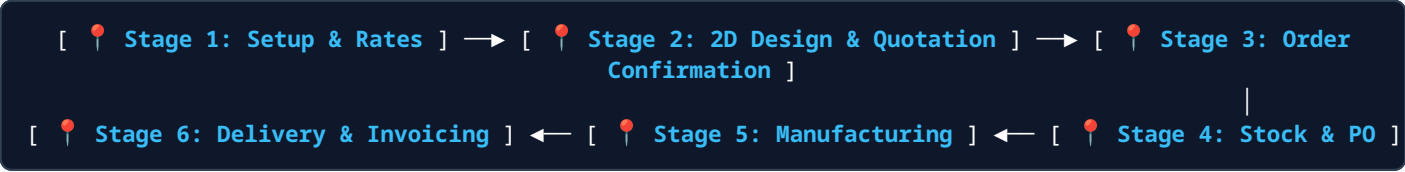


1. END-TO-END SYSTEM PROCESS WORKFLOW



Stage 1: System Setup & Master Rates Configuration

- 1. **System Branding** (/settings): Configure Company Name, Address, Contact Details, Primary Logo, and Secondary Co-branding Logo.
- 2. **Alu Database Rates** (/alu/database):
 - **Profiles Tab**: Profile codes, bar lengths (18' / 21'), cost per full bar.
 - **Glass Tab**: Glass types, thickness (5, 6, 8mm), SqFt rates, processing.
 - **Accessories Tab**: Locks, hinges, rollers, gaskets/wool felt (per meter).
 - **Applications Tab**: Window/Door formulas, BOM, Labour rate methods.

Stage 2: 2D Design & Quotation Generation

- 1. **2D Interactive Configurator** (/alu/configurator):
 - Input outer dimensions (W, H in mm) & select Product Category.
 - View real-time 2D Canvas rendering & instant cost estimations.
 - Inspect Aluminium, Glass, Hardware, Labour costs, and Waste %.
- 2. **Quotation Engine** (/alu/quotations/new):
 - Input Client & Project details. Runs 1D Bin Packing & 2D Glass Nesting.
 - Add Transport, Charges, Profit Margin % for Final Selling Price.
- 3. **Export & Submission**: Download dual-branded PDF quotation.

Stage 3: Order Confirmation & Production Release

- 1. Update Quotation Status to **Accepted / Confirmed** upon client approval.
- 2. System automatically generates:
 - **Sales Order**
 - **Production Order / Job Card**
- 3. **Auto BOM Explosion**: Compiles total project material requirements (Aluminium bars, Glass sheets, Hardware counts).

Stage 4: Stock Reservation & Purchase Orders

- 1. **Stock Availability Check**:
 - **In Stock**: Auto-reserved (StockReservation). Prioritizes reusable off-cuts (AluScrap) to minimize waste.
 - **Out of Stock**: Auto-triggers Purchase Requisition (PR) / PO.
- 2. **GRN Entry**: Log Goods Receipt Note (GRN) upon supplier delivery to update available inventory.

Stage 5: Shop Floor Manufacturing & Fabrication

- 1. **Release Workshop Diagrams**:
 - **1D Aluminium Cutting Sheet**: Visual bar nesting and offsets.
 - **2D Glass Cutting Sheet**: Optimized 2D sheet nesting layouts.
- 2. **Off-Cut Recovery**: Reusable off-cut bars are automatically registered into (AluScrap) Inventory for future jobs.

Stage 6: Dispatch, Invoicing & Profitability

- 1. **Logistics & Gate Pass** (/logistics/gate-passes): Generate dispatch gate passes and track site delivery.
- 2. **Invoicing & Payments** (/invoices , /payments): Issue customer invoices and log advance/final payments.
- 3. **P&L Analytics** (/reports/daily-pnl): Analyze real-time net margins, cost variances, and material consumption.

2. CORE TECHNICAL VARIABLE DEFINITIONS

VARIABLE SYMBOL	DESCRIPTION & SCOPE	UNIT OF MEASUREMENT
W	Window / Door Outer Frame Width	Millimeters (mm)
H	Window / Door Outer Frame Height	Millimeters (mm)
P	Number of Panels per Opening Unit (e.g., 1, 2, 3)	Count (Integer)

VARIABLE SYMBOL	DESCRIPTION & SCOPE	UNIT OF MEASUREMENT
Q	Total Quantity of Openings / Units	Count (Integer)
W_{glass}	Net Glass Panel Cut Width	Millimeters (mm)
H_{glass}	Net Glass Panel Cut Height	Millimeters (mm)



3. BOM & CUTTING DIMENSION FORMULAS (PRODUCT TEMPLATES)

♦ (A) SLIDING DOOR — 3 PANEL (2 TRACK, OXX SLIDING CONFIGURATION)

ALUMINIUM PROFILE CUT LENGTHS			GLASS SPECIFICATIONS	
PROFILE COMPONENT	CUTTING FORMULA	QTY / UNIT	Glass Width (W _{glass})	(W / 3) - 100 mm
Outer Frame Vertical (L/R)	H	2	Glass Height (H _{glass})	H - 180 mm
Outer Frame Horizontal (Top/Bot)	W	2	Total Glass Panels	3 × Q
Bottom Track Profile	W	1	HARDWARE & ACCESSORIES	
Top Track Profile	W	1	Double Rollers	6 × Q
Sash Vertical Profile	H - 50 mm	6	C-Groove Handles	2 × Q
Sash Horizontal Profile	W / 3 mm	6	Multi-point Locks	2 × Q
Interlock Vertical Profile	H - 50 mm	2	Wool Pile / Weatherstrip	[6H + 6(W/3)] / 1000 m
			Weatherproof Silicone	2 × Q Cartridges

♦ (B) CASEMENT WINDOW — 2 PANEL CASEMENT

ALUMINIUM PROFILE CUT LENGTHS			GLASS SPECIFICATIONS	
PROFILE COMPONENT	CUTTING FORMULA	QTY / UNIT	Glass Width (W _{glass})	(W / 2) - 130 mm
Outer Frame Vertical	H	2	Glass Height (H _{glass})	H - 140 mm
Outer Frame Horizontal	W	2	Total Glass Panels	2 × Q
Sash Frame Vertical	H - 40 mm	4	HARDWARE & ACCESSORIES	
Sash Frame Horizontal	(W / 2) - 30 mm	4	Friction Hinges (12")	4 × Q
Centre Mullion Profile	H - 75 mm	1	Cockspur Handles	2 × Q
Glass Beading Vertical	H - 140 mm	4	Corner Brackets	8 × Q
Glass Beading Horizontal	(W / 2) - 130 mm	4	Rubber Gasket / Beading	[4H + 4(W/2)] / 1000 m

♦ (C) FIXED GLASS — 1 PANEL

ALUMINIUM PROFILE CUT LENGTHS

PROFILE COMPONENT	CUTTING FORMULA	QTY / UNIT
Outer Frame Vertical	H	2
Outer Frame Horizontal	W	2
Glass Beading Vertical	H - 40 mm	2
Glass Beading Horizontal	W - 40 mm	2

GLASS SPECIFICATIONS

Glass Width (W_{glass})	W - 40 mm
Glass Height (H_{glass})	H - 40 mm
Total Glass Panels	1 × Q



4. OPTIMIZATION, WASTAGE & COSTING ENGINE FORMULAS

Aluminium 1D Bar Wastage %

Waste Length Calculation (mm):

Waste Length = Total Purchased Length - Total Cut Length

Wastage Percentage (%):

Wastage % = (Waste Length / Total Purchased Length) × 100

Glass Panel SqFt Area & 2D Sheet Costing

Glass Panel Area Calculation (SqFt):

Glass Area (SqFt) = [(W_{glass} × H_{glass}) / 92,903.04] × Quantity

* Unit conversion factor: 1 SqFt = 92,903.04 mm²

Glass 2D Sheet Costing Formula:

Total Glass Cost = (Purchased Glass Sheets Count × 32 SqFt) × Unit Rate

* Standard Full Sheet Dimensions = 2438 mm × 1219 mm ≈ 32 SqFt

Costing Engine & Selling Price Calculation

1. Direct Material Cost:

Material Cost = Aluminium Bar Cost + Glass Sheet Cost + Accessories Cost

2. Labour Cost Calculation Options:

Option A: Per Opening

Labour = Q × Labour Rate

Option B: Per SqFt

Labour = Total Area × Labour Rate

Option C: Percentage Method

Labour = Material Cost × (Rate % / 100)

3. Base Cost Before Profit Margin:

Base Cost = Material Cost + Labour Cost + Transport Cost + Additional Charges

4. Profit Margin & Final Client Selling Price:

Final Selling Price (Client Price) = [Base Cost + (Base Cost × Profit Margin % / 100)] - Discounts ± Manual Adjustments

⚙️ **Database & ERP Integration Note:** All system workflows, formulas, material profiles, unit rates, wastage thresholds, and accessory ratios listed above are fully integrated into **Sidebar** → **ALUECO ALUMINIUM** → **Alu Database** (/alu/database).